One Sentence Pitch

SWARM is a dual approach in response to Deformed Wing Virus and Varroa mite infestation, a virus and parasite which impact over 60% of honey-bee colonies in the United States.

Our system consists of a de-novo minibinder and RNAi system, both delivered through a sucrose solution

Abstract (150)

Abstract here! Link closely to agriculture, food insecurity, and economic motives

Defining the Problem

Problem dwv varroa here. 62% of colonies lost wtv stats

Epfl graphic with global population



Current Solutions

Current treatments and issues here

- Narroa
- Folic and oxalic acid

Our Approach

SWARM here!

Brief description of minibinders, rnai, prince and models, hardware, and sdgs

Lambert iGEM designed SWARM to protect honey bees from deformed wing virus (DWV) by targeting both the virus and the mite that spreads it, *Varroa destructor*.

We designed small proteins called minibinders to grab the virus's outer shell, and double-stranded RNA (dsRNA) to switch off genes critical to the mite's survival. Together, the two treatments target both infection and transmission.

Introduction

SWARM addresses a threat to one of agriculture's most important pollinators. Honey bees visit crop flowers more often than any other single species, and many crops rely on them heavily [1]. In our home state of Georgia, beekeepers lost about 88,000 colonies between October 2025 and June 2026, translating to ~20% of the state's colonies lost every three months [2].

Across the United States, beekeepers lost 39.9% of their managed colonies between April 2025 and April 2026 [3]. Worldwide, the share of crop production that depends directly on animal pollinators is worth an estimated $235 billion to $577 billion USD each year [4]. Parasites, pesticides, and a lack of flowers all contribute to bee declines, but most U.S. colony losses trace back to the *Varroa destructor* mite and the virus it spreads, deformed wing virus (DWV) [5],[6].

DWV threatens honey bees worldwide, and when researchers tested bees from 603 apiaries across the United States, they detected DWV in 89.4% of them [7],[8]. The mite wedges itself under the overlapping plates of a bee's abdomen, pierces the soft membrane between them, and feeds on the fat tissue underneath [9] (Figure 1). Each feeding wound gives DWV a direct route into the bee [7].

Without mites, DWV infections usually cause no visible symptoms, but mites drive viral levels high enough to cause deformed wings and early death [10],[11]. Each new round of brood adds more mites and more infected bees, and an untreated colony will likely die [12],[14].

Current treatments face two shortfalls: mites resist several existing miticides, and the Honey Bee Health Coalition's 2026 guide lists treatments for the mite only [6],[14].

Table 1. Current Varroa treatments

Approach	Examples	Limitations
Synthetic miticides	Amitraz, tau-fluvalinate, coumaphos	Mites resist them; residues persist in wax [6],[14],[15]

Organic acids	Formic acid, oxalic acid	Formic acid can kill brood in heat; oxalic acid misses mites under cappings [14]
Mechanical and cultural methods	Drone brood removal, screened bottom boards	Work poorly on their own [14]
RNA interference (registered in 2025)	Vadescana, sold as Norroa	Targets one mite gene, not the virus; needs open brood [12],[14]

As one member of the Metro Atlanta Beekeepers Association said, "It's not a matter of if there are Varroa mites. Every colony has them."

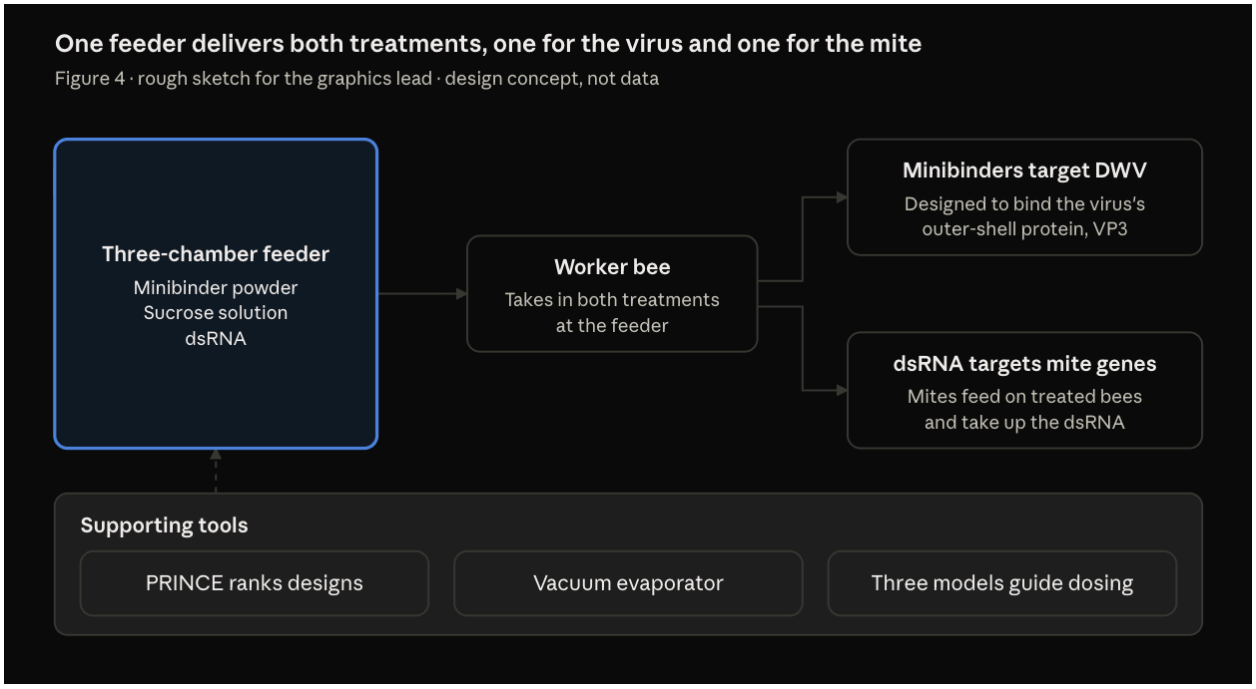
The result from lacking existing treatments and a threat that continues to pose difficulty for colonies is an ecological harm at large.

Introducing SWARM

SWARM began with four design requirements, drawn from the gaps in current treatments and from beekeepers' needs:

- Treat DWV directly, since current guidance lists only mite treatments [14].
- Reduce mites without the brood harm or wax residues of chemicals [14],[15].
- Slow resistance, because mites already resist several miticides [6],[14].
- Fit into hive routines as a practical, low-cost treatment.

To meet them, SWARM pairs two treatments in one hive-top feeder: minibinders against DWV and dsRNA against *Varroa* (Figure 3). Bees would take in both treatments through the feeder's sucrose solution. We designed the minibinders to act on the virus in the bee's gut, and the dsRNA to reach the mite when it feeds on a treated bee, a route earlier researchers demonstrated [16].



Minibinders

SWARM targets DWV with minibinders, small proteins designed on a computer to bind a specific target. We designed our minibinders against VP3, one of the proteins that form the virus's outer shell, or capsid. Part of VP3 folds into a knob on the virus's surface that may help the virus enter bee cells [17]. DWV researchers we consulted also supported VP3 as a target [add statement].

We designed the minibinders to inhibit viral replication by binding VP3 to block host cell binding (Figure 4). We compared minibinders with antibodies and with nanobodies, small antibody fragments from camels and llamas, and chose minibinders for their small size and heat tolerance (Table 2).

Table 2. Antibodies compared with minibinders and nanobodies

Property	Antibodies	Nanobodies	Minibinders

Size (in kilodaltons, a measure of protein mass)	Large, multi-part proteins of about 150 kilodaltons [18]	A single antibody domain of about 15 kilodaltons [18]	Fewer than ~100 amino acids [19]
Heat tolerance	Lose activity after heating [20]	High stability [18]	Keep activity after exposure to high temperatures [20]
Origin	Made by an animal's immune system or selected in the lab	Taken from antibodies that camels and llamas make [18]	Designed on a computer from the target's shape alone [19]

Nanobodies were the primary option we considered prior to minibinders however Dr. Aashish Manglik from UCSF advised us on the difficulty with design stating, “For our computational design projects, we still make libraries of hundreds or thousands of candidates.” Through more conversations with stakeholders and literature, we decided to pivot towards minibinder design as it had far greater promise in creating a binding molecule.

To choose which designs to test, we built PRINCE, software trained on more than 5,000 experimentally validated minibinder designs (see Software). PRINCE first finds binding hotspots, the places on VP3 most likely to anchor a binder, then ranks each design on predicted binding and predicted expression, or how much usable protein gets produced (Figure 5). Through two cycles of binder expression testing, we created the first ever minibinder for deformed wing virus!

RNAi

SWARM targets the mite with RNA interference, a natural process in which cells use dsRNA to switch off target genes. When a mite cell takes in dsRNA that matches one of its genes, the cell destroys that gene's messenger RNA (the working copy it uses to build a protein) to limit it from making that protein (Figure 6).

We designed a multiplexed dsRNA that targets three *Varroa* salivary gland genes essential to mite survival. By designing novel siRNA sequences for Vd-CHIsal, Vd-CHIBIN, and VdAV422, we are able to silence three genes compared to one in existing remedies [12]. Earlier work that silenced a single mite gene reduced mite populations by over 60%, so targeting three should not only boost efficacy but also limit the likelihood of resistance that mites are developing [16].

Our dsRNA approach is delivered through treated bees, the same route used to successfully silence mite genes in the past [16].

To address ecological concerns with reduced mite populations, we spoke to entomologist Jeff Pettis, who assured us it would be “completely fine to eliminate Varroa, as honeybees get no benefit at all from it (*see iHP–Jeff Pettis*).”

Delivery

SWARM reaches bees through a hive-top sucrose feeder. Our design dries the minibinders into a powder, intended to be shelf-stable, with a low-cost vacuum evaporator. The evaporator removes water at low temperature so heat does not unfold, or denature the proteins [NEED: whether drying has been performed, the drying temperature, the cost in U.S. dollars, stability results, and why low-temperature drying matters given the heat tolerance in Table 2].

A three-chamber feeder holds the minibinder powder, a sucrose solution, and the dsRNA in separate compartments, so one unit delivers both treatments to the colony (Figure 7).